

Creating Graphs Using PHP



Graphs can be created in several ways, ranging from manual to the use of computers and sophisticated graph-drawing software packages. This article gives a brief description of the use of PHP to draw graphs and, in particular, the JpGraph package.

There are many software packages that help create graphs and charts using PHP. One such package is JpGraph. It is released under a dual licence – the QPL 1.0 (Qt Free License) for open source or educational use, and the JpGraph Professional License for commercial use. Let's discuss the open source version of JpGraph in this article.

Installation

The package can be downloaded from <http://jpgraph.net/download/>. Currently, there are two different packages for PHP4 and PHP5. Select the appropriate link. You may extract the files anywhere, preferably in the document root (e.g., `/var/www`).

In order to use JpGraph, you will also have to download the *gd* library for PHP:

```
sudo apt-get install php5-gd
```

To check whether the library has been installed properly, type the following command:

```
php5 -m
```

You should see *gd* in the list. See Figure 1.

That's it! Now we can start creating graphs.

Creating a line graph

Let us look at how to create a simple line graph. Locate the directory where *jpgraph.php* can be found. Include this file in your code:

```
require_once('jpgraph/src/jpgraph.php');
```

Next, in order to draw a line graph, you need to include the *jpgraph_line.php* file, as follows:

```
require_once('jpgraph/src/jpgraph_line.php');
```

Now, to create a graph, instantiate the Graph class by passing its width and the height, respectively.

```
$graph = new Graph(300, 400); //300 is the width and 400 is the height
$graph->SetScale('intln'); //set the scale of the graph
```

Prepare the data to be plotted as a line graph using an array: